

Question 1

$$\text{Let } f(x) = \begin{cases} x^2 - 3x - 4 & \text{if } x < 0 \\ -\cos x - 4 & \text{if } x > 0 \end{cases}$$

Calculate (i) $\lim_{x \rightarrow 0^-} f(x)$ (ii) $\lim_{x \rightarrow 0^+} f(x)$ (iii) $\lim_{x \rightarrow 0} f(x)$

Question 2

$$\text{Let } f(x) = \frac{10 - x^2}{x - 5}$$

Calculate (i) $\lim_{x \rightarrow 0} f(x)$ (ii) $\lim_{x \rightarrow 5^+} f(x)$ (iii) $\lim_{x \rightarrow \infty} f(x)$

Question 3

$$\text{Let } f(x) = \frac{x - 8}{x^2 - 64}$$

Calculate (i) $\lim_{x \rightarrow 8^-} f(x)$ (ii) $\lim_{x \rightarrow 8^+} f(x)$ (iii) $\lim_{x \rightarrow \infty} f(x)$

Question 4

$$\text{Let } f(x) = \frac{x - 10}{x^2 + 2x + 1}$$

Calculate (i) $\lim_{x \rightarrow -1^-} f(x)$ (ii) $\lim_{x \rightarrow -1^+} f(x)$ (iii) $\lim_{x \rightarrow -1} f(x)$

Question 5

Calculate

i $\lim_{x \rightarrow 0} \sin x$ ii $\lim_{x \rightarrow 0} \cos x$ iii $\lim_{x \rightarrow 0^-} \operatorname{cosec} x$ iv $\lim_{x \rightarrow 0^+} \cot x$
 v $\lim_{x \rightarrow \pi/2} \sin x$ vi $\lim_{x \rightarrow \pi/2} \cos x$ vii $\lim_{x \rightarrow \pi/2^+} \sec x$ viii $\lim_{x \rightarrow \pi/2^+} \tan x$

Question 6

Calculate

i $\lim_{x \rightarrow \pi} \frac{\sin x}{x}$ ii $\lim_{x \rightarrow \pi} \frac{\sin(x^2 - \pi^2)}{x - \pi}$ iii $\lim_{x \rightarrow \pi} \frac{\sin(x - \pi)}{x^2 - \pi^2}$ iv $\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 6x}$

Question 7

Calculate (i) $\lim_{x \rightarrow \infty} \frac{1 - 4x^2}{2x^2 + 4x - 6}$ (ii) $\lim_{x \rightarrow -\infty} \frac{1 - 4x^3}{2x^2 + 4x - 6}$ (iii) $\lim_{x \rightarrow \infty} \frac{1 - 4x^5}{2x^6 + 4x - 6}$

Question 8

Calculate (i) $\lim_{x \rightarrow 1^+} \frac{\sqrt{x} - 1}{\sqrt{x} - 1}$ (ii) $\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x + 21} - 5}$ (iii) $\lim_{x \rightarrow 3} \frac{3 - x}{\frac{1}{x} - \frac{1}{3}}$

Question 9

Calculate (i) $\lim_{x \rightarrow \infty} e^{-3x}$ (ii) $\lim_{x \rightarrow -\infty} e^{3x}$ (iii) $\lim_{x \rightarrow \infty} e^{5x}$ (iv) $\lim_{x \rightarrow \infty} e^{1/x}$

Question 10

Calculate (i) $\lim_{x \rightarrow 1} \ln x$ (ii) $\lim_{x \rightarrow \infty} \ln x$ (iii) $\lim_{x \rightarrow 0^-} \ln x$ (iv) $\lim_{x \rightarrow 2^+} \ln(x - 2)$

Question 11

Calculate (i) $\lim_{x \rightarrow 0} \arctan x$ (ii) $\lim_{x \rightarrow \infty} \arctan(2x - 3)$ (iii) $\lim_{x \rightarrow -\infty} \arctan(-2x + 3)$

Question 12

Calculate (i) $\lim_{x \rightarrow 1} \frac{|x| - 1}{x - 1}$ (ii) $\lim_{x \rightarrow 0} \frac{|x| - 1}{x - 1}$ (iii) $\lim_{x \rightarrow 1^-} \frac{|x - 1|}{x - 1}$

Question 13

Calculate (i) $\lim_{x \rightarrow 0} \frac{1}{x^2}$ (ii) $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x^2}\right)$ (iii) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x^2}\right)$

Question 14

Calculate (i) $\lim_{x \rightarrow -\infty} \frac{e^{-x}}{e^x - 1}$ (ii) $\lim_{x \rightarrow -\infty} \frac{e^x}{e^{-x} - 1}$ (iii) $\lim_{x \rightarrow \infty} \frac{e^{2x}}{e^{3x} - 1}$